

The Exact Spectral Gap of Kac's Master Equation: A Complete Conditional-Projection Induction

HCS-C322 theorem-and-evidence package

3 September 2026 — revision round 1

Abstract

We give a normalization-locked proof of the exact gap for Kac's uniform binary-collision master equation. On $S^{N-1}(\sqrt{N})$, with unordered-pair angle average Q_N and positive generator $L_N = N(I - Q_N)$, the gap is $(N+2)/[2(N-1)]$. We reproduce the infinite-dimensional lower induction via coordinate conditional expectations, identify the quartic equality mode and its multiplicity, and close the $N = 2$, positive-energy, and zero-energy boundaries. Exact polynomial matrices audit the algebra but are not offered as the lower-bound proof.

1 Frozen collision operator

Let σ_N be normalized surface measure on $X_N = S^{N-1}(\sqrt{N})$, $N \geq 2$. If $R_{ij}(\theta)$ rotates the (i, j) coordinate plane, define

$$Q_N f(v) = \binom{N}{2}^{-1} \sum_{i < j} \frac{1}{2\pi} \int_{-\pi}^{\pi} f(R_{ij}(\theta)v) d\theta, \quad L_N = N(I - Q_N). \quad (1)$$

Thus pairs are unordered, the angular measure is a probability, and the factor N belongs to the generator. Kac introduced this collision model in kinetic theory [1]; the exact gap architecture below is due to Carlen, Carvalho, and Loss [2].

Each pair average is the orthogonal projection onto the functions invariant under its circle action. Hence Q_N is a positive self-adjoint Markov contraction. Its fixed functions are constants: coordinate-plane rotations generate $SO(N)$, which is transitive on X_N . Put

$$\lambda_N = \sup_{\substack{\|f\|_2=1 \\ \langle f, 1 \rangle=0}} \langle f, Q_N f \rangle, \quad \Delta_N = N(1 - \lambda_N).$$

Let

$$F_N(v) = \sum_{i=1}^N v_i^4 - \frac{3N^2}{N+2}.$$

The centering uses $\mathbb{E}_{\sigma_N} v_i^4 = 3N/(N+2)$. Direct angle averaging gives

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \{(u \cos \theta + v \sin \theta)^4 + (-u \sin \theta + v \cos \theta)^4\} d\theta = \frac{3}{4}(u^2 + v^2)^2. \quad (2)$$

Summing (2) over pairs and using $\sum_i v_i^2 = N$ yields

$$Q_N F_N = \left(1 - \frac{N+2}{2N(N-1)}\right) F_N, \quad L_N F_N = \frac{N+2}{2(N-1)} F_N. \quad (3)$$

This is an upper bound for the gap, not yet a proof that no slower mode exists.

2 The conditional-projection lower bound

For $P_j f = \mathbb{E}(f \mid v_j)$ set $P = N^{-1} \sum_j P_j$. Fixing $v_j = y$ leaves an $(N-1)$ -coordinate sphere. Let $Q_{N-1}^{(j)}$ be the pair average excluding j , with the remaining sphere rescaled if needed. Counting each pair in exactly $N-2$ slices gives

$$\langle f, Q_N f \rangle = \frac{1}{N} \sum_{j=1}^N \int \langle f_{j,y}, Q_{N-1}^{(j)} f_{j,y} \rangle d\nu_N(y). \quad (4)$$

Equivalently, if $L_{N-1}^{(j)} = (N-1)(I - Q_{N-1}^{(j)})$, then $L_N = (N-1)^{-1} \sum_j L_{N-1}^{(j)}$ as quadratic forms. Centering every slice by $P_j f(y)$ in (4) and applying λ_{N-1} gives, for unit mean-zero f ,

$$\langle f, Q_N f \rangle \leq \lambda_{N-1} + (1 - \lambda_{N-1}) \langle f, P f \rangle. \quad (5)$$

The exact norm of P is therefore decisive.

Lemma 1 (Coordinate-projection spectrum). *For $N \geq 3$, on constants-perpendicular $L^2(\sigma_N)$,*

$$\|P\| = \mu_N = \frac{N+4}{N(N+1)}.$$

Its top eigenspace is one-dimensional and is generated by F_N .

Proof. On one-coordinate functions define the self-adjoint conditional operator $(K_N g)(x) = \mathbb{E}[g(v_2) \mid v_1 = x]$. Conditional spherical moments give

$$K_N(x^{2r}) = c_{r,N}(N - x^2)^r, \quad c_{r,N} = \frac{(2r - 1)!!}{(N - 1)(N + 1) \cdots (N + 2r - 3)}, \quad (6)$$

and symmetry kills odd functions. Even polynomial spaces are invariant. Self-adjointness and density of polynomials for the compact coordinate marginal produce a complete orthogonal-polynomial eigenbasis, whose eigenvalues are the leading coefficients in (6):

$$\alpha_0 = 1, \quad \alpha_{2r} = (-1)^r c_{r,N}, \quad \alpha_{2r+1} = 0.$$

Since $|\alpha_{2r+2}/\alpha_{2r}| = (2r + 1)/(N + 2r - 1) < 1$, the largest mean-zero eigenvalue is $\kappa_N = \alpha_4 = 3/(N^2 - 1)$ and the smallest is $\alpha_2 = -1/(N - 1)$.

We now transfer this spectrum without presuming that P has only point spectrum. Put $H_0 = L_0^2(\nu_N)$ and define $T : H_0^N \rightarrow L_0^2(\sigma_N)$ by $T(h_1, \dots, h_N) = \sum_j h_j(v_j)$. Then $T^*f = (P_1 f, \dots, P_N f)$ and $P = TT^*/N$. The nonzero spectra of TT^*/N and T^*T/N agree: besides the usual eigenvector transfer, for $z \neq 0$ the resolvent identity

$$(TT^* - z)^{-1} = -z^{-1}I + z^{-1}T(T^*T - z)^{-1}T^*$$

(and its adjoint counterpart) rules out unaccounted continuous or approximate spectrum. The block matrix of T^*T/N has diagonal I/N and off-diagonal K_N/N . On a K_N eigenfunction of eigenvalue α , the trivial and standard coordinate-index subspaces therefore have eigenvalues

$$\frac{1 + (N - 1)\alpha}{N}, \quad \frac{1 - \alpha}{N}.$$

The complete polynomial resolution above makes these blocks exhaustive; their only accumulation is $1/N$, already the infinite-multiplicity odd block. Any additional point for P is zero from $\ker T^*$. Consequently the largest mean-zero spectral value is

$$\max \left\{ \frac{1 + (N - 1)\kappa_N}{N}, \frac{1 - \alpha_2}{N} \right\} = \frac{N + 4}{N(N + 1)}.$$

The second entry is $1/(N - 1)$, strictly smaller for $N > 2$. Simplicity of α_4 and of the trivial index line, followed by T , identifies the unique top vector with the sum of centered coordinate quartics, namely F_N . This proves both the norm and uniqueness. $\square$

Theorem 1 (Exact gap, multiplicity, and decay). *For every $N \geq 2$,*

$$\text{gap}(L_N) = \Delta_N = \frac{N + 2}{2(N - 1)}.$$

For $N \geq 3$ the slow eigenspace is $\text{span}\{F_N\}$. At $N = 2$, every mean-zero function is a slow eigenfunction. For every mean-zero f and $t \geq 0$,

$$\|e^{-tL_N} f\|_2 \leq e^{-t\Delta_N} \|f\|_2,$$

and the rate is sharp.

Proof. Taking the supremum in (5) and applying Lemma 1,

$$\begin{aligned} \Delta_N &\geq \frac{N}{N - 1}(1 - \mu_N)\Delta_{N-1} \\ &= \left(1 - \frac{3}{N^2 - 1}\right)\Delta_{N-1} = \frac{(N - 2)(N + 2)}{(N - 1)(N + 1)}\Delta_{N-1}. \end{aligned}$$

For $N = 2$, the sole pair is averaged over the entire circle, so Q_2 is projection onto constants and $\Delta_2 = 2$. Therefore

$$\Delta_N \geq 2 \prod_{j=3}^N \frac{(j - 2)(j + 2)}{(j - 1)(j + 1)} = 2 \frac{N + 2}{4(N - 1)} = \frac{N + 2}{2(N - 1)}.$$

Equation (3) attains this lower bound. For $N \geq 3$, equality in the induction forces equality in the P variational bound; Lemma 1 then gives the one-dimensional slow eigenspace. The $N = 2$ multiplicity follows from $Q_2 = 0$ on the whole mean-zero sector. The spectral theorem for the nonnegative self-adjoint L_N gives the semigroup bound, and the stated eigenspaces make it sharp. $\square$

Energy boundary. On $S^{N-1}(\sqrt{E})$ for any $E > 0$, dilation by $\sqrt{E/N}$ unitarily conjugates the walk above and leaves the gap unchanged; the centered quartic is $\sum_i v_i^4 - 3E^2/(N + 2)$, since $\mathbb{E}v_i^4 = 3E^2/[N(N + 2)]$. At $E = 0$ the state space is one point and the mean-zero Hilbert space is $\{0\}$, so no positive relaxation gap is assigned.

Revision certificate: complete conditional-projection gap proof

This round adds the one-coordinate spectrum, exact projection norm, telescoping lower induction, multiplicity, decay, and energy boundaries.

References

- [1] M. Kac, “Foundations of kinetic theory,” in *Proc. Third Berkeley Symp. Math. Statist. Probab.*, vol. III (1956), 171–197, UC Berkeley Library record 112857.
- [2] E. A. Carlen, M. C. Carvalho, and M. Loss, “Determination of the spectral gap for Kac’s master equation and related stochastic evolution,” *Acta Math.* 191 (2003), DOI: 10.1007/BF02392695; arXiv:math-ph/0109003.